Generative AI for Finance
Spring 2026

RICE | BUSINESS
Jones Graduate School of Business

Instructor

Kerry Back
kerryback@gmail.com
J. Howard Creekmore Professor of Finance and Professor of Economics

Meeting Schedule

TTh 12:30 – 2:00
3/17/2026 – 4/23/2026

Special session:

- Saturday, March 21, 9:00-12:00 - optional Python session

All sessions will be in McNair 212, including the special session.

Course Description

The course is about financial applications of generative AI. We will discuss and use current AI tools but also discuss broader concepts of AI and how they relate to finance. The course will be "hands-on." We will be working in class on laptops each day.

We will use Anthropic's Claude throughout the course. Anthropic has made a number of important innovations that the rest of the industry has followed, including a protocol for connecting AI to tools (Model Context Protocol), coding agents that can execute terminal commands (Claude Code), and code execution in a virtual machine (Claude Cowork). Claude has also generally led all other models in coding tasks for the past two years. We will also briefly look at ChatGPT, Gemini, and other large language models.

Learning Objectives

1. Use AI coding agents (Claude Code) to perform financial analysis, build visualizations, and generate reports
2. Apply AI to core finance tasks: portfolio optimization, DCF valuation, and sentiment-based trading signals
3. Connect AI to external data and documents via MCP servers, skills, and retrieval-augmented generation
4. Automate multi-step financial workflows with reusable skills, reporting pipelines, and AI agents
5. Verify and govern AI-generated analysis using red-teaming, audit protocols, and structured review

Grading

Grades will be based on six group assignments (15% each) and peer assessments (10%). Groups can consist of no more than five students. Each assignment consists of multiple exercises and will take a nontrivial amount of time to complete. Do not wait until the last minute to get started.

The assignments are posted on Canvas. An assignment is due each Tuesday at 11:59 pm beginning March 24 and ending April 28 (during exam week).

Claude Accounts

Each student will be reimbursed for a Claude Pro account ($20 per month). The Pro account provides access to Cowork, Claude Code, and the Claude plug-in for Excel.

Slide Decks

1. Overview
2. Exploring Claude
3. Mean-Variance Analysis
4. Claude Desktop and Claude Code
5. Sub-Agents and Skills
6. Makers and Checkers
7. Some Claude Tips
8. Web Scraping with AI
9. Chatbots and Agents
10. Corporate AI Deployment
11. Magnolia DCF
12. Retrieval Augmented Generation
13. Building Agents with the Claude API
14. Sentiment Analysis

Honor Code

The Rice University honor code applies to all work in this course.

Disability Accommodations

Any student with a documented disability requiring accommodations in this course is encouraged to contact me outside of class. All discussions will remain confidential. Any adjustments or accommodations regarding assignments or the final exam must be made in advance. Students with disabilities should also contact Disability Support Services in the Allen Center.